Newman-Penrose Formalism Calculations

Stephani et al., Equation (35.77)

Metric

$$ds^2 = \left(-t^a x^{-\frac{2\,b^2}{a^2+b^2-2\,a-1}+\frac{4\,a}{a^2+b^2-2\,a-1}+\frac{2}{a^2+b^2-2\,a-1}}\right)\mathrm{d}t\otimes\mathrm{d}t + \left(t^a x^{-\frac{2\,b^2}{a^2+b^2-2\,a-1}+\frac{4\,a}{a^2+b^2-2\,a-1}+\frac{2}{a^2+b^2-2\,a-1}}\right)\mathrm{d}x\otimes\mathrm{d}x$$
$$+\left(t^{-b+1} x^{-\frac{2\,b^2}{a^2+b^2-2\,a-1}+\frac{4\,a}{a^2+b^2-2\,a-1}+\frac{2}{a^2+b^2-2\,a-1}}\right)\mathrm{d}y\otimes\mathrm{d}y + \left(t^{b+1} x^{-\frac{2\,b^2}{a^2+b^2-2\,a-1}+\frac{4\,a}{a^2+b^2-2\,a-1}+\frac{2}{a^2+b^2-2\,a-1}}\right)\mathrm{d}z\otimes\mathrm{d}z$$

Coordinates

$(x^\mu) = (t, x, y, z)$

Null Tetrad

Covariant components

$$l = \left(\frac{\sqrt{\frac{1}{2}}\sqrt{t^a}\,x^{\frac{a^2}{a^2+b^2-2\,a-1}}}{x}\right)dt + \left(\frac{\sqrt{\frac{1}{2}}\sqrt{t^a}\,x^{\frac{a^2}{a^2+b^2-2\,a-1}}}{x}\right)dx$$

$$n = \left(\frac{\sqrt{\frac{1}{2}}\sqrt{t^a}\,x^{\frac{a^2}{a^2+b^2-2\,a-1}}}{x}\right)dt + \left(-\frac{\sqrt{\frac{1}{2}}\sqrt{t^a}\,x^{\frac{a^2}{a^2+b^2-2\,a-1}}}{x}\right)dx$$

$$m = \left(\frac{\sqrt{\frac{1}{2}}\sqrt{t}\sqrt{t^a}\,x^{\frac{a^2}{a^2+b^2-2\,a-1}}}{t^{\frac{1}{2}\,a}t^{\frac{1}{2}\,b}x}\right)dy + \left(\frac{i\,\sqrt{\frac{1}{2}}\sqrt{t}\sqrt{t^a}\,t^{\frac{1}{2}\,b}x^{\frac{a^2}{a^2+b^2-2\,a-1}}}{t^{\frac{1}{2}\,a}x}\right)dz$$

$$\overline{m} = \left(\frac{\sqrt{\frac{1}{2}}\sqrt{t}\sqrt{t^a}\,x^{\frac{a^2}{a^2+b^2-2\,a-1}}}{t^{\frac{1}{2}\,a}t^{\frac{1}{2}\,b}x}\right)dy + \left(-\frac{i\,\sqrt{\frac{1}{2}}\sqrt{t}\sqrt{t^a}\,t^{\frac{1}{2}\,b}x^{\frac{a^2}{a^2+b^2-2\,a-1}}}{t^{\frac{1}{2}\,a}x}\right)dz$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}\,x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}\right)\,dt + \left(\frac{\sqrt{2}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}\,x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}\right)\,dx$$

$$n = \left(-\frac{\sqrt{2}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}\,x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}\right)\,dt + \left(-\frac{\sqrt{2}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}\,x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}\right)\,dx$$

$$m = \left(\frac{\sqrt{2}\sqrt{t^a}t^{\frac{1}{2}\,b}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,\sqrt{t}t^{\frac{1}{2}\,a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}\,x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}\right)\,dy + \left(\frac{i\,\sqrt{2}\sqrt{t^a}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,\sqrt{t}t^{\frac{1}{2}\,a}t^{\frac{1}{2}\,b}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}\,x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}\right)\,dz$$

$$\overline{m} = \left(\frac{\sqrt{2}\sqrt{t^a}t^{\frac{1}{2}\,b}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,\sqrt{t}t^{\frac{1}{2}\,a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}\,x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}\right)\,dy + \left(-\frac{i\,\sqrt{2}\sqrt{t^a}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,\sqrt{t}t^{\frac{1}{2}\,a}t^{\frac{1}{2}\,b}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}\,x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}\right)\,dz$$

Directional Derivatives

$$D = -\frac{\sqrt{2}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}\,x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}\,\partial_t + \frac{\sqrt{2}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}\,x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}\,\partial_x$$

$$\Delta = -\frac{\sqrt{2}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}\,x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}\,\partial_t - \frac{\sqrt{2}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}\,x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}\,\partial_x$$

$$\delta = \frac{\sqrt{2}\sqrt{t^a}t^{\frac{1}{2}\,b}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,\sqrt{t}t^{\frac{1}{2}\,a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}\,x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}\,\partial_y + \frac{i\,\sqrt{2}\sqrt{t^a}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,\sqrt{t}t^{\frac{1}{2}\,a}t^{\frac{1}{2}\,b}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}\,x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}\,\partial_z$$

$$\overline{\delta} = \frac{\sqrt{2}\sqrt{t^a}t^{\frac{1}{2}\,b}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,\sqrt{t}t^{\frac{1}{2}\,a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}\,x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}\,\partial_y - \frac{i\,\sqrt{2}\sqrt{t^a}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,\sqrt{t}t^{\frac{1}{2}\,a}t^{\frac{1}{2}\,b}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}\,x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}\,\partial_z$$

Spin Coefficients

$$\rho = \frac{\sqrt{2}b^2}{2\,(a^2+b^2-2\,a-1)\sqrt{t^a}x^{\frac{a^2}{a^2+b^2-2\,a-1}}} - \frac{\sqrt{2}a}{(a^2+b^2-2\,a-1)\sqrt{t^a}x^{\frac{a^2}{a^2+b^2-2\,a-1}}} + \frac{\sqrt{2}a^2x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{4\,(a^2+b^2-2\,a-1)t\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}$$
$$+ \frac{\sqrt{2}b^2x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{4\,(a^2+b^2-2\,a-1)t\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}} - \frac{\sqrt{2}}{2\,(a^2+b^2-2\,a-1)\sqrt{t^a}x^{\frac{a^2}{a^2+b^2-2\,a-1}}}$$
$$- \frac{\sqrt{2}ax^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,(a^2+b^2-2\,a-1)t\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}} - \frac{\sqrt{2}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{4\,(a^2+b^2-2\,a-1)t\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}$$

$$\sigma = -\frac{\sqrt{2}bx^{\frac{b^2}{a^2+b^2-2\,a-1}}}{4\,t\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}$$

$$\kappa = 0$$

$$\mu = \frac{\sqrt{2}b^2}{2\,(a^2+b^2-2\,a-1)\sqrt{t^a}x^{\frac{a^2}{a^2+b^2-2\,a-1}}} - \frac{\sqrt{2}a}{(a^2+b^2-2\,a-1)\sqrt{t^a}x^{\frac{a^2}{a^2+b^2-2\,a-1}}} - \frac{\sqrt{2}a^2x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{4\,(a^2+b^2-2\,a-1)t\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}$$
$$- \frac{\sqrt{2}b^2x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{4\,(a^2+b^2-2\,a-1)t\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}} - \frac{\sqrt{2}}{2\,(a^2+b^2-2\,a-1)\sqrt{t^a}x^{\frac{a^2}{a^2+b^2-2\,a-1}}}$$
$$+ \frac{\sqrt{2}ax^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,(a^2+b^2-2\,a-1)t\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}} + \frac{\sqrt{2}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{4\,(a^2+b^2-2\,a-1)t\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}$$

$$\lambda = \frac{\sqrt{2}bx^{\frac{b^2}{a^2+b^2-2\,a-1}}}{4\,t\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{\sqrt{2}b^2}{4\,(a^2+b^2-2\,a-1)\sqrt{t^a}x^{\frac{a^2}{a^2+b^2-2\,a-1}}} - \frac{\sqrt{2}a^3x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{8\,(a^2+b^2-2\,a-1)t\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}} - \frac{\sqrt{2}ab^2x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{8\,(a^2+b^2-2\,a-1)t\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}$$
$$+ \frac{\sqrt{2}a}{2\,(a^2+b^2-2\,a-1)\sqrt{t^a}x^{\frac{a^2}{a^2+b^2-2\,a-1}}} + \frac{\sqrt{2}a^2x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{4\,(a^2+b^2-2\,a-1)t\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}$$
$$+ \frac{\sqrt{2}}{4\,(a^2+b^2-2\,a-1)\sqrt{t^a}x^{\frac{a^2}{a^2+b^2-2\,a-1}}} + \frac{\sqrt{2}ax^{\frac{b^2}{a^2+b^2-2\,a-1}}}{8\,(a^2+b^2-2\,a-1)t\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}$$

$$\gamma = -\frac{\sqrt{2}b^2}{4\,(a^2+b^2-2\,a-1)\sqrt{t^a}x^{\frac{a^2}{a^2+b^2-2\,a-1}}} + \frac{\sqrt{2}a^3x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{8\,(a^2+b^2-2\,a-1)t\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}} + \frac{\sqrt{2}ab^2x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{8\,(a^2+b^2-2\,a-1)t\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}$$
$$+ \frac{\sqrt{2}a}{2\,(a^2+b^2-2\,a-1)\sqrt{t^a}x^{\frac{a^2}{a^2+b^2-2\,a-1}}} - \frac{\sqrt{2}a^2x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{4\,(a^2+b^2-2\,a-1)t\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}$$
$$+ \frac{\sqrt{2}}{4\,(a^2+b^2-2\,a-1)\sqrt{t^a}x^{\frac{a^2}{a^2+b^2-2\,a-1}}} - \frac{\sqrt{2}ax^{\frac{b^2}{a^2+b^2-2\,a-1}}}{8\,(a^2+b^2-2\,a-1)t\sqrt{t^a}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}$$

Ricci Tensor Components

$$
\begin{aligned}
\Phi_{00} = & -\frac{a^4 b^2 x^2}{8\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} - \frac{a^2 b^4 x^2}{4\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
& - \frac{b^6 x^2}{8\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} + \frac{a^5 x^2}{4\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
& + \frac{a^3 b^2 x^2}{(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} + \frac{3\,ab^4 x^2}{4\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
& - \frac{a^2 b^2}{2\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} - \frac{7\,a^4 x^2}{8\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
& - \frac{a^2 b^2 x^2}{(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} + \frac{3\,b^4 x^2}{8\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
& + \frac{a^3 b^2 x^{\frac{a^2}{a^2+b^2-2\,a-1}} x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)tt^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}} x^{\frac{2\,a}{a^2+b^2-2\,a-1}} x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}} \\
& + \frac{ab^4 x^{\frac{a^2}{a^2+b^2-2\,a-1}} x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)tt^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}} x^{\frac{2\,a}{a^2+b^2-2\,a-1}} x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}} + \frac{a^3}{(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
& - \frac{3\,ab^2 x^2}{2\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} - \frac{a^4 x^{\frac{a^2}{a^2+b^2-2\,a-1}} x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)tt^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}} x^{\frac{2\,a}{a^2+b^2-2\,a-1}} x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}} \\
& - \frac{2\,a^2 b^2 x^{\frac{a^2}{a^2+b^2-2\,a-1}} x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)tt^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}} x^{\frac{2\,a}{a^2+b^2-2\,a-1}} x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}} + \frac{a^2}{2\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
& + \frac{5\,a^2 x^2}{4\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} - \frac{3\,b^2 x^2}{8\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
& + \frac{3\,a^3 x^{\frac{a^2}{a^2+b^2-2\,a-1}} x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)tt^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}} x^{\frac{2\,a}{a^2+b^2-2\,a-1}} x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}} \\
& - \frac{ab^2 x^{\frac{a^2}{a^2+b^2-2\,a-1}} x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)tt^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}} x^{\frac{2\,a}{a^2+b^2-2\,a-1}} x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}} \\
& + \frac{3\,ax^2}{4\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} + \frac{2\,a^2 x^{\frac{a^2}{a^2+b^2-2\,a-1}} x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)tt^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}} x^{\frac{2\,a}{a^2+b^2-2\,a-1}} x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}} \\
& + \frac{x^2}{8\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} + \frac{ax^{\frac{a^2}{a^2+b^2-2\,a-1}} x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)tt^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}} x^{\frac{2\,a}{a^2+b^2-2\,a-1}} x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}
\end{aligned}
$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$\Phi_{10} = 0$

$$\Phi_{11} = -\frac{a^4b^2x^2}{16\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^ax^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} - \frac{a^2b^4x^2}{8\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^ax^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}}$$
$$- \frac{b^6x^2}{16\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^ax^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} + \frac{a^5x^2}{8\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^ax^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}}$$
$$+ \frac{a^3b^2x^2}{2\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^ax^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} + \frac{3\,ab^4x^2}{8\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^ax^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}}$$
$$+ \frac{a^2b^2}{4\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^ax^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} - \frac{7\,a^4x^2}{16\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^ax^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}}$$
$$- \frac{a^2b^2x^2}{2\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^ax^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} + \frac{3\,b^4x^2}{16\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^ax^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}}$$
$$- \frac{a^3}{2\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^ax^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} - \frac{3\,ab^2x^2}{4\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^ax^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}}$$
$$- \frac{a^2}{4\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^ax^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} + \frac{5\,a^2x^2}{8\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^ax^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}}$$
$$- \frac{3\,b^2x^2}{16\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^ax^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} + \frac{3\,ax^2}{8\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^ax^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}}$$
$$+ \frac{x^2}{16\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^ax^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}}$$

$\Phi_{12} = 0$

$\Phi_{20} = 0$

$\Phi_{21} = 0$

$$
\begin{aligned}
\Phi_{22} = &-\frac{a^4b^2x^2}{8\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} - \frac{a^2b^4x^2}{4\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
&- \frac{b^6x^2}{8\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} + \frac{a^5x^2}{4\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
&+ \frac{a^3b^2x^2}{(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} + \frac{3\,ab^4x^2}{4\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
&- \frac{a^2b^2}{2\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} - \frac{7\,a^4x^2}{8\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
&- \frac{a^2b^2x^2}{(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} + \frac{3\,b^4x^2}{8\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
&- \frac{a^3b^2x^{\frac{a^2}{a^2+b^2-2\,a-1}}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)tt^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}} \\
&- \frac{ab^4x^{\frac{a^2}{a^2+b^2-2\,a-1}}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)tt^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}} + \frac{a^3}{(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
&- \frac{3\,ab^2x^2}{2\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} + \frac{a^4x^{\frac{a^2}{a^2+b^2-2\,a-1}}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)tt^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}} \\
&+ \frac{2\,a^2b^2x^{\frac{a^2}{a^2+b^2-2\,a-1}}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)tt^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}} + \frac{a^2}{2\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
&+ \frac{5\,a^2x^2}{4\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} - \frac{3\,b^2x^2}{8\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
&- \frac{3\,a^3x^{\frac{a^2}{a^2+b^2-2\,a-1}}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)tt^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}} \\
&+ \frac{ab^2x^{\frac{a^2}{a^2+b^2-2\,a-1}}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)tt^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}} \\
&+ \frac{3\,ax^2}{4\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} - \frac{2\,a^2x^{\frac{a^2}{a^2+b^2-2\,a-1}}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)tt^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}} \\
&+ \frac{x^2}{8\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} - \frac{ax^{\frac{a^2}{a^2+b^2-2\,a-1}}x^{\frac{b^2}{a^2+b^2-2\,a-1}}}{2\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)tt^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}x^{\frac{2\,a}{a^2+b^2-2\,a-1}}x^{\left(\frac{1}{a^2+b^2-2\,a-1}\right)}}
\end{aligned}
$$

$$
\begin{aligned}
\Lambda = {} & \frac{a^4 b^2 x^2}{48\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} + \frac{a^2 b^4 x^2}{24\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
& + \frac{b^6 x^2}{48\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} - \frac{a^5 x^2}{24\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
& - \frac{a^3 b^2 x^2}{6\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} - \frac{ab^4 x^2}{8\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
& - \frac{a^2 b^2}{4\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} - \frac{b^4}{2\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
& + \frac{7\,a^4 x^2}{48\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} + \frac{a^2 b^2 x^2}{6\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
& - \frac{b^4 x^2}{16\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} + \frac{a^3}{2\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
& + \frac{2\,ab^2}{(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} + \frac{ab^2 x^2}{4\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
& - \frac{7\,a^2}{4\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} + \frac{b^2}{(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
& - \frac{5\,a^2 x^2}{24\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} + \frac{b^2 x^2}{16\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
& - \frac{2\,a}{(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} - \frac{ax^2}{8\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} \\
& - \frac{1}{2\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}} - \frac{x^2}{48\,(a^4+2\,a^2b^2+b^4-4\,a^3-4\,ab^2+2\,a^2-2\,b^2+4\,a+1)t^2t^a x^{\frac{2\,a^2}{a^2+b^2-2\,a-1}}}
\end{aligned}
$$

Weyl Tensor Components

$$
\Psi_0 = -\frac{abx^{\frac{2\,b^2}{a^2+b^2-2\,a-1}}}{4\,t^2t^a x^{\frac{4\,a}{a^2+b^2-2\,a-1}} x^{\frac{2}{a^2+b^2-2\,a-1}}}
$$

$$
\Psi_1 = 0
$$

$$
\Psi_2 = \frac{b^2 x^{\frac{2\,b^2}{a^2+b^2-2\,a-1}}}{12\,t^2t^a x^{\frac{4\,a}{a^2+b^2-2\,a-1}} x^{\frac{2}{a^2+b^2-2\,a-1}}} + \frac{ax^{\frac{2\,b^2}{a^2+b^2-2\,a-1}}}{12\,t^2t^a x^{\frac{4\,a}{a^2+b^2-2\,a-1}} x^{\frac{2}{a^2+b^2-2\,a-1}}} - \frac{x^{\frac{2\,b^2}{a^2+b^2-2\,a-1}}}{12\,t^2t^a x^{\frac{4\,a}{a^2+b^2-2\,a-1}} x^{\frac{2}{a^2+b^2-2\,a-1}}}
$$

$\Psi_3 = 0$

$$\Psi_4 = -\frac{abx^{\frac{2\,b^2}{a^2+b^2-2\,a-1}}}{4\,t^2 t^a x^{\frac{4\,a}{a^2+b^2-2\,a-1}}\,x^{\frac{2}{a^2+b^2-2\,a-1}}}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"